

Trends and Controlling Factors of Vegetation Change in Different Regions of China

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Abstract: Understanding the complex relationship between vegetation change and both natural and anthropogenic factors is a subject of global importance. However, comprehensive explanations of vegetation cover trends across China's different regions and the dynamic roles of their drivers remain limited. This study analyzed national and regional vegetation change trends from 2000 to 2020 and evaluated the evolving impacts of natural and anthropogenic factors. Results indicate that 44.14% of China's land experienced significant increase ($P < 0.05$) in vegetation coverage. The Northeast (A1), Southwest (A5), and South China (A8) regions showed extremely significant increases in vegetation cover, with over 65% of vegetation exhibiting extremely significant growth ($P < 0.01$). In contrast, less than 25% of vegetation in Inner Mongolia (A2), Northwest (A3), and the Qinghai-Tibetan Plateau (A4) subregions demonstrated an extremely significant increasing trend ($P < 0.01$). Precipitation ($q = 0.766$) and land use type ($q = 0.636$) were the most influential natural and anthropogenic factors, respectively, with their interaction ($q = 0.838$) dominating national vegetation patterns. On the west side of the Hu Line, vegetation dynamics were mainly limited by arid and semi-arid climates, with precipitation as the dominant factor, though land use measures have contributed to some vegetation improvement. Between 2000 and 2020, the influence of precipitation on vegetation cover increased in regions A3 and A4, with q -values rising by 26.73% and 101.13%, respectively. Additionally, soil type exerted a significant effect ($P < 0.001$) on vegetation cover across all regions, being most pronounced in A2 ($q = 0.692$). On the east side of the Hu Line, vegetation growth benefited generally from warm and humid conditions, while local decline in urbanized areas was largely attributable to land use change and economic expansion. Concurrently anthropogenic factors such as land use and population density increasingly influenced vegetation dynamics in eastern urban areas of the Hu Line. Population density and GDP were the most influential factors affecting vegetation cover in region A8, with q -values of 0.443 and 0.380, respectively ($P < 0.001$). Future efforts should maintain the benefits of large-scale ecological projects and harmonize the relationship between urban vegetation and anthropogenic influences.

Keywords: vegetation change; climate change; Normalized Difference Vegetation Index (NDVI); anthropogenic factor; GeoDetector

Citation: LIU Yuan, WANG Chengyuan, WANG Enheng, MAO Xuegang, FENG Tiantian, YANG Liwei, NIU Qian, DING Longxin, SERGEY Chumachenko, 2025. Trends and Controlling Factors of Vegetation Change in Different Regions of China. *Chinese Geographical Science*, 35(6): 1269–1282. <https://doi.org/10.1007/s11769-025-1568-4>; <https://cstr.cn/32177.14.cgs/s11769-025-1568-4>

Received date: 2024-12-20; accepted date: 2025-05-16

Foundation item: Under the auspices of the National Natural Science Foundation of China (No. 32371863), Fundamental Research Funds for the Central Universities (No. 2572025AW39)

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1 Introduction

Vegetation is the primary component of terrestrial ecosystems and it is necessary for the supply of ecosystem services, such as soil preservation, climate management, hydrological processes, carbon cycling and maintaining ecosystem stability (Piao et al., 2011; Forzieri et al., 2017; Zhang et al., 2018). Vegetation is also a sensitive indicator of ecosystem reactions to alterations in the climate and to human-caused disruptions (Fernandes et al., 2016). Therefore, the monitoring and attribution of changes in vegetation dynamics have become key issues in responding to changes in the global system in recent decades (Piao et al., 2011). Furthermore, studying vegetation dynamics contributes to a better understanding of the interactions between surface processes, climate activities, human activities and vegetation to support sustainable development processes in terrestrial ecosystems (Angelini et al., 2011; Pudmenzky et al., 2015; Zhu et al., 2020).

From 2000 to 2017, China accounted for only 6.6% of the world's total vegetation coverage, yet contributed 25% of the global net increase in leaf area (Chen et al., 2019; Piao et al., 2020; Su et al., 2020). More than 40.00% of China's vegetated areas have experienced significant changes owing to intensive human activities and unprecedented climate change (Gao et al., 2019; Su et al., 2020). The trend of vegetation cover growth in China is the most pronounced in the world, accounting for 25.00% of the global net increase in anthropogenic greening (Chen et al., 2019). In particular, 25.30% of the vegetation greening in China and 39.53% of the increase in vegetation cover in the northern region can be attributed to large-scale ecological restoration projects, including the Three-North Protective Forest Project (Zhai et al., 2023), the Returning Cultivated Land to Forestry Project (Li et al., 2022), the Beijing-Tianjin-Hebei Wind and Sand Source Management Project (Li et al., 2023) and the Natural Forest Protection Project (Fan et al., 2024). However, as a country characterized by diverse geographic environments and a plethora of land use types, the extent to which vegetation changes will vary in future across different regions, under the combined influence of human activities and global climate change, remains unclear in China, necessitating urgent further research.

Spatial and temporal changes in vegetation cover reflect complex processes that are influenced by a combination of human activities and natural factors (Piao et al., 2020). Topographic conditions, rising atmospheric CO₂ concentrations, changes in temperature and precipitation patterns and nitrogen deposition are considered to be the main natural causes of vegetation change (Piao et al., 2011; Liu et al., 2023). In addition, vegetation is often negatively affected by anthropogenic factors such as population growth, road construction, land use and economic expansion (Zhu et al., 2016; Qu et al., 2018; Yang et al., 2021). In the past, scholars have explored the characteristics, trends and causes of vegetation changes in China's Qinghai-Tibetan Plateau (Liu et al., 2019), Southwest China (Liu et al., 2020) and Loess Plateau (Nie et al., 2021). Statistical methods such as partial correlation, residual analysis and regression analysis were used to study the linear relationship between vegetation changes and driving causes and to understand the underlying mechanisms (Leroux et al., 2017; Gu et al., 2018; Qu et al., 2018). However, there is no complete explanation as to whether the significant differences in vegetation change in different regions of China are dominated by natural or anthropogenic factors, or whether the coupled influence of the two characterizes the trend of vegetation change. In addition, the persistent influence of natural and anthropogenic factors on the trends of vegetation change and their spatial distribution in different regions of China over time needs to be further investigated.

This study divided China into eight regions for detailed study. The analysis was based on Normalized Difference Vegetation Index (NDVI) data from 2000 to 2020. The goals of the research were to identify the alterations in vegetation brought about by human and natural drivers, as well as to comprehend the mechanisms that caused these alterations. More precisely, this study centered on three questions: 1) what was China's vegetation cover distribution and trend between 2000 and 2020? 2) Which natural and human causes contributed to the development of these trends and how were their interactions demonstrated or displayed? 3) What are the distinct characteristics of natural and anthropogenic impacts on vegetation over time? This study contributes to future regional vegetation restoration and management in China.

2 Materials and Methods

2.1 Study area

Study area covers all of China, including Hong Kong, Macao, and Taiwan. China's climatic range and complex geography led to the division of the country into eight zones for further study (Fig. 1). The subregions of Inner Mongolia (A2) and Northwest (A3) belong to the arid and semiarid inland zones; the Qinghai-Tibetan Plateau (A4) is a unique alpine and independent region; and the Northeast (A1), Southwest (A5), North China (A6), the Middle and Lower Yangtze River (A7) and South China (A8) are typical humid and semi-humid monsoon zones (Qu et al., 2015). The Hong Kong, Macao, and Taiwan of China were classified as South China (A8).

2.2 Data sources and pre-processing

The data sources used in this study from 2000 to 2020 are shown in Table 1. This study used the NDVI as a metric for vegetation assessment. To mitigate the impact of undesirable low-value noise, such as cloud and rain, the Maximum Value Composite (MVC) method was used to combine the NDVI data from 2000 to 2020 (Nie et al., 2021). We chose three types of factors to study: topography, climate and human activities. Topographic characteristics such as elevation, slope, aspect and soil type exert a substantial influence on vegetation distribution and collectively contribute to the spatial diversity of vegetation patterns (Guo et al., 2014). The Digital Elevation Model (DEM) and soil type data employed follow the update cycles of the Chinese

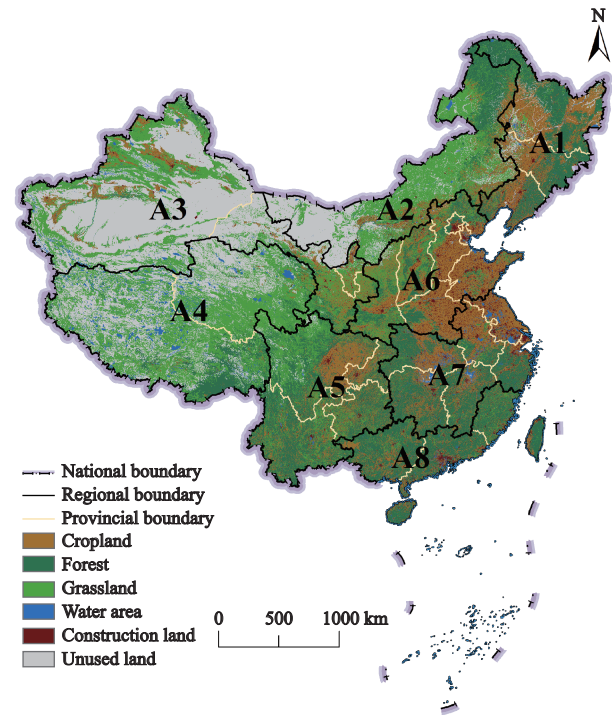


Fig. 1 The geographical arrangement of China's A1–A8 areas and land use distribution of 2020 (<https://www.resdc.cn/>). Based on the standard map service website of the National Platform for Common GeoSpatial Information Services (<https://www.tianditu.gov.cn/>) with the approval number GS(2024)0650, the boundary of the base map has not been modified. A1: Northeast; A2: Inner Mongolia; A3: Northwest; A4: Qinghai-Tibetan Plateau; A5: Southwest; A6: North China; A7: the Middle and Lower Yangtze River; A8: South China

Academy of Sciences Geospatial Data Cloud and the Resources and Environmental Science Data Center, respectively. Temperature and precipitation are also strongly linked to changes in vegetation conditions and

Table 1 Sources of data for this research

Type	Data description	Resolution	Source
Vegetation cover	NDVI	1 km	https://www.resdc.cn , March 18, 2023
Topographic	DEM (X_1) / m	90 m	https://www.gscloud.cn , March 20, 2023
	Slope (X_2) / (°)	90 m	Derived from DEM data
	Aspect (X_3)	90 m	Derived from DEM data
	Soil type (X_4)	1 km	https://www.resdc.cn , March 28, 2023
Climate	Precipitation (X_5) / (mm)	1 km	https://data.cma.cn , March 6, 2023
	Temperature (X_6) / °C	1 km	https://data.cma.cn , March 6, 2023
Anthropogenic	Land use (X_7)	1 km	https://www.resdc.cn , March 6, 2023
	Population density (X_8) / (people/km ²)	1 km	https://landscan.ornl.gov , March 18, 2023
	GDP Population density (X_9) / million yuan (RMB)	1 km	https://www.resdc.cn , March 18, 2023

Notes: NDVI, Normalized Difference Vegetation Index. GDP, Gross Domestic Product. Precipitation and temperature data were spatially interpolated using Inverse Distance Weighting from station records to monthly resolutions and subsequently aggregated to annual timescales

are thought to be the main elements determining the spatial pattern of vegetation among all the drivers that may impact plant development (Georganos et al., 2017). Human factors like Gross Domestic Product (GDP), population density and land use also affect how vegetation is distributed spatially (Nie et al., 2021). To maintain data consistency, all data were first transformed to the same projection, administratively masked and resampled to a final uniform resolution of 1000 m.

Second, the GeoDetector model used in this study can only handle discrete variables (Wang et al., 2016). Therefore, all selected factors were converted to a discrete format (Fig. S1). To determine the NDVI thresholds for classifying vegetated and non-vegetated areas, we referred to previous studies (Jabal et al., 2022) and combined them with the actual distribution trend of unutilized land (Fig. 1). We classified NDVI into five distinct categories: non-vegetation cover [0, 0.2), low vegetation cover [0.2, 0.4), medium vegetation cover [0.4, 0.6), greater vegetation cover [0.6, 0.8) and high vegetation cover [0.8, 1.0]. Elevation, precipitation and temperature were categorized into nine classes using the natural breakpoint method in ArcGIS 10.7. Slope and land use types were categorized into six classes based on the Technical Specification for Land Use Status Survey (Nie et al., 2021). Using existing specifications, slope orientation was categorized into nine categories. The soil types were reclassified into 21 groups using the People's Republic of China's 1 : 1 000 000 soil map as a guide. Significant geographical variation was present in the 2020 spatial distributions of the nine factors (Fig. S1).

2.3 Methods

2.3.1 Trend analysis

Thiel-Sen trend analysis (Theil, 1992) and the Mann-Kendall test (Mann, 1945; Kendall, 1949) were used to assess the vegetation change from 2000 and 2020, which are commonly used for trend analysis of remotely sensed time series (Sun et al., 2022). They compensate for the shortcomings of traditional linear methods that are susceptible to outlier data or large measurement errors (Liu et al., 2010) and are extensively employed in detecting vegetation changes (Liang et al., 2020). The calculation method for the Thiel-Sen slope (β) is presented in Theil (1992). When the slope $\beta > 0$, it indicates a tendency toward growth; when $\beta =$

0, it indicates a trend without change and when the slope $\beta < 0$, it shows a declining trajectory.

Trend tests were conducted at a given α significance level using a two-sided trend test (Table 2). This work presents three significance levels: $\alpha = 0.01$, $\alpha = 0.05$ and $\alpha = 0.10$. With 90%, 95% and 99% confidence levels, the statistic Z was used to conduct a trend test, the trend passed the significance test if $|Z|$ was greater than 1.65, 1.96 and 2.58, respectively.

2.3.2 GeoDetector

The GeoDetector assesses spatial coherence to identify spatial variations and measure the influence of drivers (Wang et al., 2016). This study employs factor detector and interactive detector models to analyze the effects of topography, climate, and anthropogenic factors on vegetation cover change from 2000 to 2020.

3 Results and Analyses

3.1 Trends in vegetation cover in China and in different regions

From 2000 to 2020, the vegetation cover (NDVI) in China increased at a rate of 0.0028/yr ($P < 0.05$) (Fig. 2a). Approximately 58.59% of the vegetation in China showed an increasing trend, of which 36.43% and 7.71% showed extremely significant ($P < 0.01$) and significant increasing trends ($P < 0.05$), respectively (Fig. 2b). At the regional level, there were significant differences in the trends of vegetation change. The Hu Line roughly corresponds to the 400 mm isohyet, marking the boundary between the semi-humid and semi-arid zones (Ruan et al., 2022). Therefore, this study analyzes vegetation cover changes across eight regions using the

Table 2 Dynamic degree of vegetation greening

β	Z	Trend type	Trend features
$\beta > 0$	$ Z > 2.58$	4	Extremely significant increase
	$1.96 < Z \leq 2.58$	3	Significant increase
	$1.65 < Z \leq 1.96$	2	Slightly significant increase
	$ Z \leq 1.65$	1	No significant increase
$\beta = 0$		0	No change
$\beta < 0$	$ Z \leq 1.65$	-1	No significant reduction
	$1.65 < Z \leq 1.96$	-2	Slightly significant reduction
	$1.96 < Z \leq 2.58$	-3	Significant reduction
	$ Z > 2.58$	-4	Extremely significant reduction

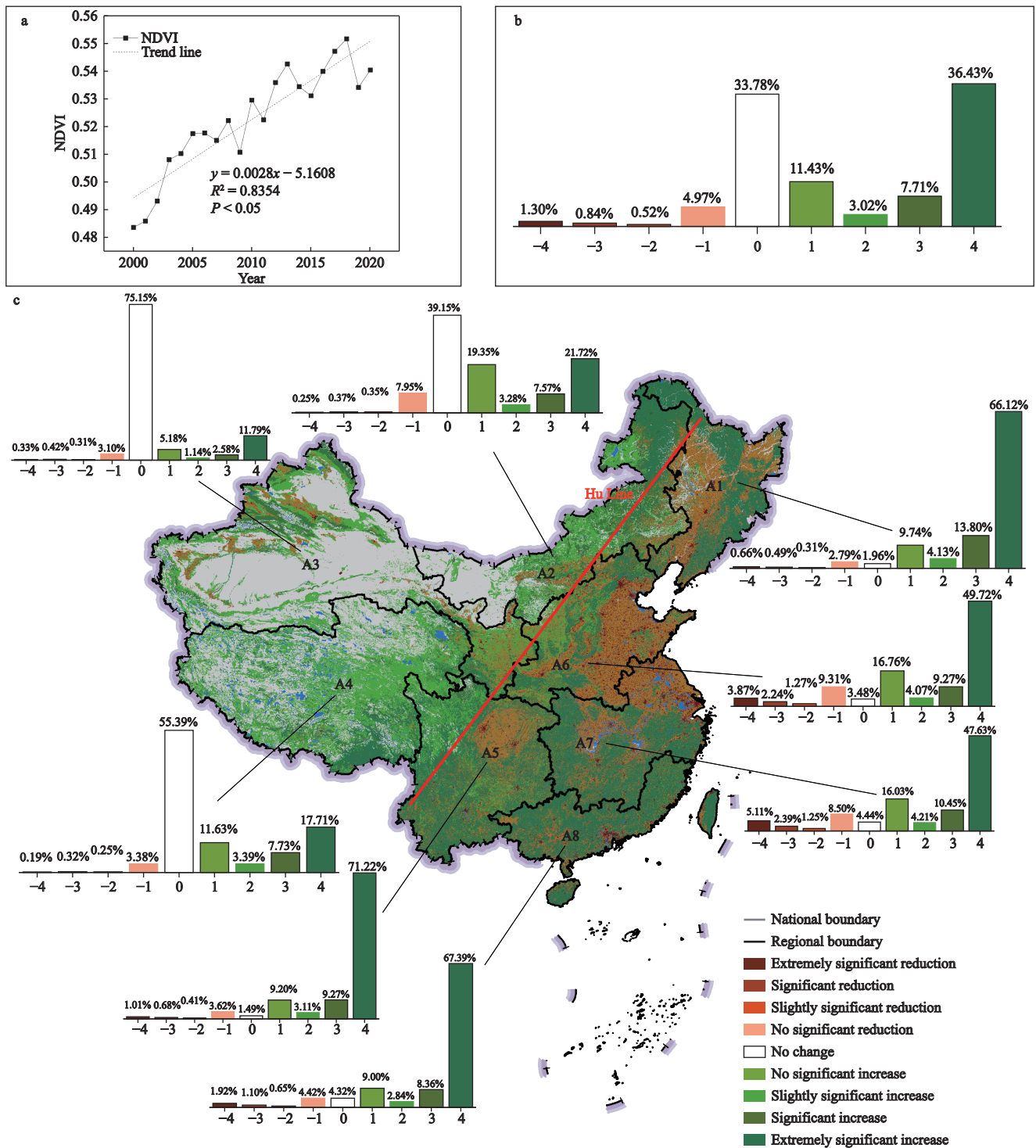


Fig. 2 Trends in vegetation change in China from 2000–2020. a. Normalized Difference Vegetation Index (NDVI) temporal trend of vegetation; b. overall trend of vegetation; c. vegetation trend in China's various regions. Based on the standard map service website of the National Platform for Common GeoSpatial Information Services (<https://www.tianditu.gov.cn/>) with the approval number GS(2024)0650, the boundary of the base map has not been modified. A1–A8 see Fig. 1. Meanings of numbers in the x-axis of Figs. 1b and 1c see Table 2

Hu Line as a reference. The most substantial increase in vegetation was observed in areas A1, A5 and A8, where more than 65% of the vegetation showed extremely sig-

nificant increase ($P < 0.01$) (Fig. 2c). In areas A6 and A7, 49.72% and 47.63% of the vegetation, respectively, showed an extremely significant increase ($P < 0.01$). In

contrast, A2, A3 and A4 had less than 25% of the area with extremely significant increasing trend of vegetation ($P < 0.01$). The most substantial decreasing trend of vegetation was concentrated in A7 (Fig. 2c), where 5.11% of the vegetation showed an extremely significant reducing trend ($P < 0.01$) and 2.39% of the vegetation showed a significant reducing trend ($P < 0.05$), mainly in the central and eastern urbanized areas. An estimated 3.87% of the vegetation in A6 showed an extremely significant reducing trend ($P < 0.01$), which was mainly in the arable land vegetation type. In A8, vegetation exhibited extremely significant ($P < 0.01$) and significant reducing ($P < 0.05$) trend at 1.92% and 1.10%, concentrated in the coastal urbanized area. By contrast, areas A2, A3 and A4 had less than 1% of their area with an extremely significant reducing trend in vegetation ($P < 0.01$). The interaction between cropland and urbanization may have caused significant differences in vegetation degradation between 2000 and 2020. Therefore, we must be vigilant about the possibility of further vegetation degradation in areas A6 and A7.

3.2 Vegetation response to natural and anthropogenic factors

We analyzed the effects of natural and anthropogenic factors on vegetation using the 2000–2020 NDVI annual mean as an input variable. At the national scale, the average q -values (q indicates the extent to which the driver explains the change in NDVI) of natural and anthropogenic factors affecting vegetation in China were precipitation ($X5$, 0.766), land use type ($X7$, 0.636), soil type ($X4$, 0.458), population density ($X8$, 0.422), eleva-

tion ($X1$, 0.295), temperature ($X6$, 0.275), GDP ($X9$, 0.141), slope ($X2$, 0.093) and aspect ($X3$, 0.005) (Table 3). Precipitation ($X5$) was the natural factor dominating vegetation change in China. The response of vegetation cover to natural factors varied considerably in different regions. The significant increasing trend of vegetation cover in areas A3 and A4 on the west side of the Hu Line was largely influenced by precipitation ($X5$), with q -values of 0.511 and 0.591, respectively. By contrast, the response of vegetation cover to natural factors, especially precipitation, was relatively low in regions A6, A7 and A8—east side of the Hu Line. The effects of precipitation ($X5$) on vegetation cover in regions A6, A7 and A8 were all less than 0.1. Soil type ($X4$) had a significant effect on vegetation cover in each area ($P < 0.001$), with the greatest effect in area A2, with a q -value of 0.692. Altitude ($X1$) had the greatest effect (0.318) on vegetation cover in area A5, which is influenced by plateaus, mountains and basins.

The results of the study showed that land use ($X7$) had a significant effect on the regional distribution of vegetation cover ($P < 0.001$) (Table 3). The vegetation cover in areas A2 and A3 on the west side of the Hu Line was most significantly affected by land use ($X7$) ($P < 0.001$), with q -values of 0.590 and 0.542, respectively. The Three-North Protective Forest Project and the Ant Forest Project, which were implemented in China from 2000 to 2020 and increased the density of vegetation by planting trees and grasslands, played a key role in improving the vegetation cover of A2 and A3, especially in slowing down the desertification of these areas (Cai et al., 2022). Rapid urban expansion in some re-

Table 3 The ability of a single factor (q) to explain vegetation cover in China from 2000 to 2020

Area	$X1$	$X2$	$X3$	$X4$	$X5$	$X6$	$X7$	$X8$	$X9$
A1	0.112***	0.114***	0.037	0.221***	0.165***	0.186***	0.191***	0.209***	0.155*
A2	0.355***	0.167***	0.009***	0.692***	0.667***	0.459***	0.590***	0.064***	0.024**
A3	0.140***	0.124***	0.003*	0.387***	0.511***	0.130***	0.542***	0.418***	0.140***
A4	0.276***	0.163***	0.006***	0.239***	0.591***	0.226***	0.328***	0.165***	0.007***
A5	0.318***	0.066***	0.004	0.143***	0.111***	0.230***	0.203***	0.085***	0.049***
A6	0.024*	0.134***	0.006	0.180***	0.090***	0.020*	0.167***	0.122***	0.144***
A7	0.164***	0.273***	0.118***	0.281***	0.059***	0.070*	0.319***	0.288***	0.224***
A8	0.163***	0.302***	0.027	0.227***	0.044**	0.186***	0.278***	0.443***	0.380***
China	0.295***	0.093***	0.005***	0.458***	0.766***	0.275***	0.636***	0.422***	0.141***

Notes: '***': $P < 0.001$; '**': $P < 0.01$; '*': $P < 0.05$. A1–A8 see Fig. 1. $X1$ – $X9$ see Table 1

regions on the east side of the Hu Line led to the loss of agricultural land, grasslands and forests, and the changes to regional landscape patterns and land use structures affected the distribution of vegetation cover. Our results showed that population density (X_8) and GDP (X_9) were the most significant factors affecting vegetation cover in A8 ($P < 0.001$), with q -values of 0.443 and 0.380, respectively. Vegetation cover in A6 and A7 was also significantly affected by land use (X_7), population density (X_8) and GDP (X_9) ($P < 0.001$). In particular, the q -values of land use (X_7) and population density (X_8) on vegetation cover in A7 reached 0.319 and 0.288, respectively.

3.3 Impact of interactions between natural and anthropogenic factors on vegetation

The interaction of all factors produced a greater effect on vegetation cover than any one variable alone (Fig. S2), indicating that the factor interactions were all bivariate-enhanced or nonlinearly enhanced. On a national scale, the interaction between precipitation (X_5) and land use (X_7) dominated the distribution of vegetation cover in China ($q(X_5 \cap X_7) = 0.838$) (Fig. 3a). There were differences in the response of different regions to the interaction between natural and anthropogenic factors. Our results indicated that the interactions of precipitation (X_5) and land use (X_7) were all very significant for vegeta-

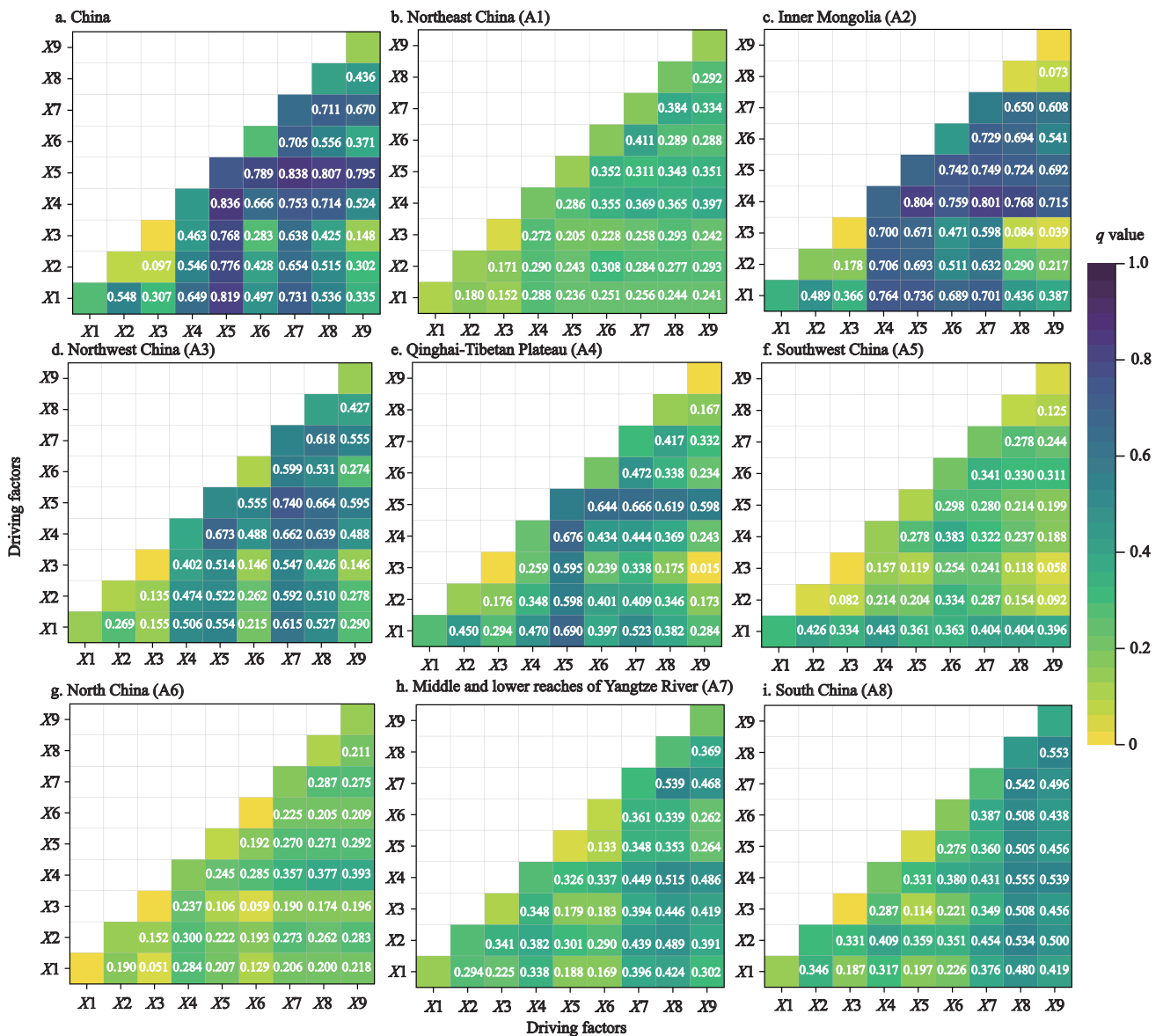


Fig. 3 The explanatory power (q -values) of interactions between drivers in eight regions (A1–A8) of China in 2020. Interaction types see Fig. S2. X_1 – X_9 see Table 1

tion cover in regions A2, A3 and A4 on the west side of the Hu Line ($P < 0.001$). The interaction between land use ($X7$) and population density ($X8$) in the A3 area also had a significant effect on vegetation cover ($P < 0.001$) ($q(X7 \cap X8) = 0.618$), suggesting that past large-scale restoration projects may have improved vegetation cover in this area. In addition, A4 was one of the most vulnerable areas to climate change, which meant that the effect of precipitation on vegetation cover may be more significant at higher elevations ($q(X1 \cap X5) = 0.690$). Regarding anthropogenic factors, we found that the interaction between land use ($X7$) and population density ($X8$) had the most significant effect on vegetation cover and the interaction effect was greatest in A7 ($P < 0.001$) ($q(X7 \cap X8) = 0.539$). Thus, anthropogenic activities,

such as rapid urbanization, affected the growth of the vegetation in A7 and led to an extremely significant reducing trend in vegetation cover ($P < 0.01$).

3.4 Dynamics of natural and anthropogenic impacts on vegetation

We used a factor detector to explore the dynamics of vegetation cover in response to each factor from 2000 to 2020. Among the natural factors, elevation ($X1$) had less fluctuating effects on vegetation cover in areas A1, A2, A3 and A4 (Figs. 4a–4d). The effect of precipitation ($X5$) on vegetation cover in A3 and A4 became larger, with q -values increasing by 26.73% and 101.13%, respectively. This suggested that precipitation ($X5$) greatly promoted the growth of grassland vegetation in A2, A3

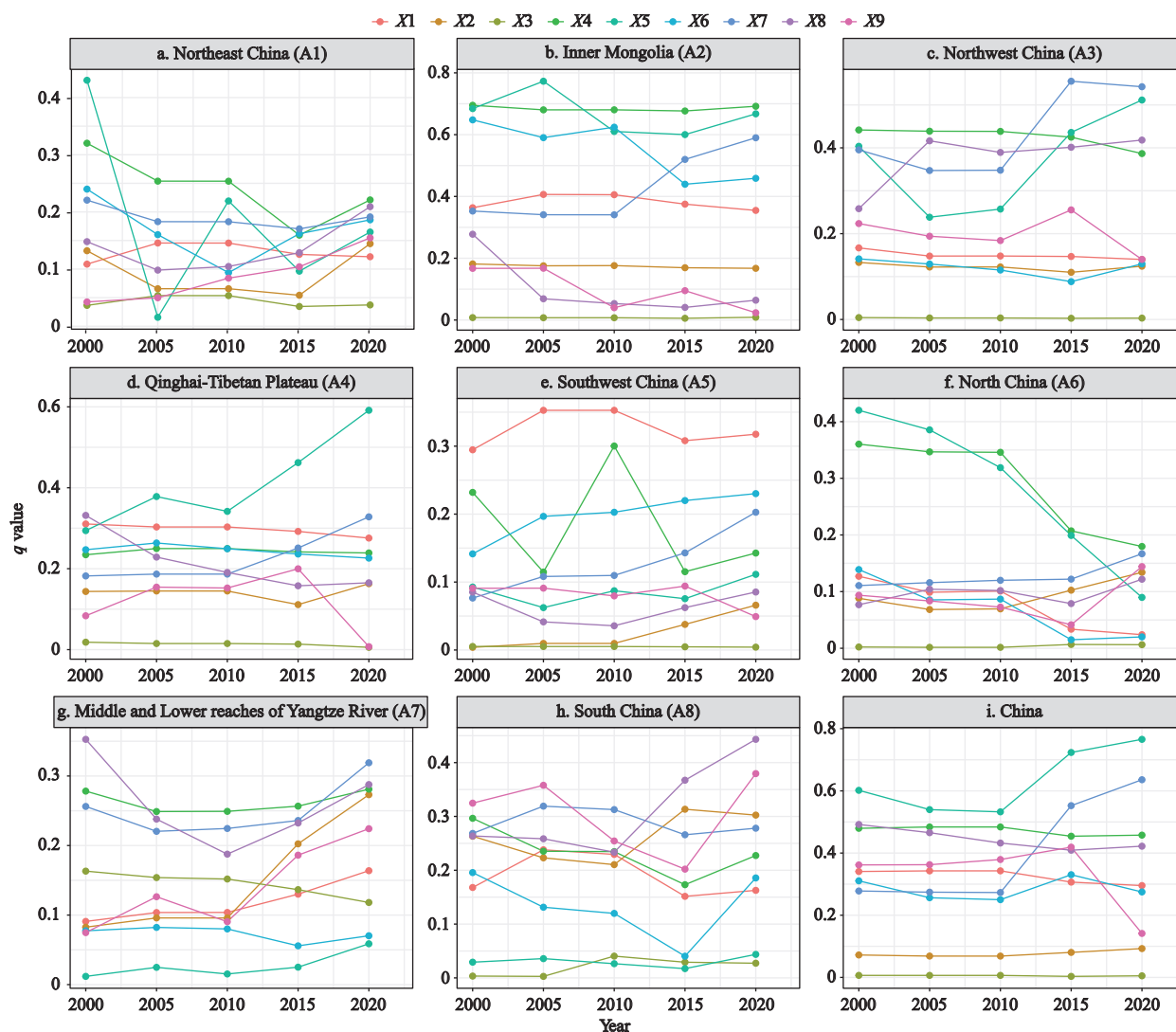


Fig. 4 The explanatory power (q -values) of drivers in explaining vegetation cover in China and subregions, 2000–2020. $X1$ – $X9$ see Table 1

and A4 from 2000 to 2020. In addition, the effect of land use type ($X7$) on vegetation was enhanced, such that the q -values of A2 and A3 land use ($X7$) were significantly enhanced by 67.26% and 37.22%, respectively. This suggests that land use ($X7$) may have had a positive impact on vegetation growth in A2 and A3 and that a series of ecological restoration projects, such as the Three-North Sheltering Forest Program and the Ant Forest Program, focusing on the A2 and A3 areas, had made progress in vegetation restoration in the past (Xiu et al., 2021; Cai et al., 2022). The influence of anthropogenic factors on vegetation cover in areas A6, A7 and A8 increased from 2000 to 2020 (Figs. 4f–4h). The influence of land use types ($X7$) on vegetation cover in A6 and A7 showed an increasing trend, with the q value increasing by 50.73% and 24.51%, respectively. The q value of population density ($X8$) on vegetation cover in A8 increased by 68.08%. Intensive human activities in A6, A7 and A8 from 2000 to 2020 led to the conversion of some arable land into urban land, which resulted in a reduction of some of the vegetation cover.

4 Discussion

4.1 Vegetation change trends along both sides of the Hu Line in China

Previous studies showed a significant vegetation growth trend in China from 2000 to 2017 (Qu et al., 2015; Chen et al., 2019). The present study extends and strengthens the pattern analysis of vegetation development trends. We found that the areas exhibiting a significant vegetation increase ($P < 0.05$) were all distributed on the eastern side of the Hu Line. And the proportion of areas with significant increasing trends were all 50% and above, while the proportion on the west side of the Hu Line remains relatively limited, all below 25% (Fig. 2). The reasons behind the significant difference in vegetation growth between the two sides of the Hu Line are related to many factors.

The impacts of climate change have contributed to significant differences in vegetation growth in different regions on both sides of the Hu Line (Xie et al., 2016; He et al., 2021). We concur with Wei et al. (2022) that the effect of sustained precipitation on vegetation in the A4 region has steadily increased (q -value increased by 101%) from 2000 to 2020. The principal factor that maintains the distribution of vegetation is precipitation,

and the Hu Line demarcates the boundary between the wet and dry zones (Ruan et al., 2022). The arid and semi-arid conditions in areas A2, A3 and A4 on the western side of the Hu Line, where the average annual precipitation is less than 400 mm (Li et al., 2024), are one of the major reasons for the significant differences in the vegetation growth patterns between the two sides of the Hu Line. The eastern side of the Hu Line has a warm and humid climate, which favors the growth of vegetation, especially subtropical broad-leaved evergreen forests (Ren et al., 2023). The climate on the western side of the Hu Line is relatively cold and dry and the vegetation growth area is mainly dominated by plateaus, mountains and basins, which limits a significant increase in vegetation to some extent (Zheng et al., 2022). In addition, there are differences in the adaptability of vegetation types to climate and environment between the two sides of the Hu Line. The eastern China has a variety of vegetation types, including forests, grasslands and farmlands, which are more adaptable to climate and environment (Yang et al., 2024). The vegetation types in the area west of the Hu Line are more homogeneous, dominated by grasslands and deserts, which are better adapted to the arid and semi-arid climate, but their growth rate and coverage are relatively low (Yuan et al., 2024). Furthermore, owing to the earlier start of economic development, areas on the eastern side of the Hu Line became aware of the need for ecological environmental protection earlier and a large number of fallow fields and forests and grasslands projects were started after 2000, which led to a more significant increase in vegetation cover (Tong et al., 2016; Liu et al., 2020; Zhang et al., 2022). The western side of the Hu Line had a later start in economic development, but in recent years, with the national attention and investment in the western region, the gradual increase in ecological restoration projects has begun to contribute to the gradual increase in vegetation (Cai et al., 2020). However, due to the ecological conditions, the survival rate of the vegetation on the west side of the Hu Line is relatively low and it is difficult to restore the vegetation (Liu et al., 2024).

In summary, the reasons for the differences in vegetation growth between the east and west sides of the Hu Line include climatic conditions, topography, human activities, vegetation adaptation and ecological policies. These factors together affect the growth and distribu-

tion pattern of vegetation on the east and west sides of the Hu Line.

4.2 Are natural or anthropogenic factors causing significant changes in vegetation?

Some previous studies have argued that the increase in precipitation in recent decades has driven the general greening of desert grasslands in northern China (Zhou et al., 2014). In this study, we similarly found that precipitation (X_5) had an increasing effect on vegetation in A3 (26% increase in q -value from 2000 to 2020). We found that land use type (X_7) also had an important influence on vegetation distribution in almost all areas, especially on the west side of the Hu Line (A2, A3 and A4), where large-scale implementation of national ecological restoration projects has led to a significant increase in vegetation cover (Cai et al., 2020).

Areas on the east side of the Hu Line (specifically A1, A5, A6, A7, and A8) exhibit a significant increasing trend ($P < 0.05$), all exceeding 50%. Previous studies have shown that climate is the main factor influencing vegetation growth in the A5 region (Liu et al., 2020). Similarly, this study found that 71.22% of area A5 exhibited an extremely significant increasing trend ($P < 0.01$), both temperature (X_6) (q -value increased by 62.66%) and precipitation (X_5) (q -value increased by 20.15%) exerted greater impacts on vegetation for the past 20 yr. However, 3.87% and 5.41% of the area in A6 and A7, respectively, showed extremely significant vegetation reduction ($P < 0.01$), which was mainly concentrated in the central and eastern regions (Fig. 2). This stems from the land urbanization process in these areas, which may affect the growth of vegetation and change the land use from arable land to urban construction land (Cao et al., 2020; Yi et al., 2021). Meanwhile, rapid urbanization, economic expansion and expansion of arable land in the eastern section of A7 may have negatively affected vegetation growth in this area (Yin et al., 2018; Shen et al., 2023). A6 and A7 are areas of high population and concentrated economic activity and these factors have a substantial impact on vegetation (Table 3). In 2020, NDVI decreased in the A6 and A8 regions as population density (X_8) increased at discrete categories (Fig. S3). Despite the apparent positive trend in vegetation growth in most areas on the east side of the Hu Line, land urbanization still requires effective government control, landscape planning and other inputs to

regulate and coordinate vegetation conservation and restoration in urban areas.

We found that the interactive effects of climate and human activities dominated the significant changes in vegetation cover in the regions. Climatic factors had the largest effect on the trend of increasing vegetation. However, negative anthropogenic activities dominated the significant decline of vegetation in some areas in the eastern side of the Hu Line. The implementation of large-scale ecological restoration projects may have contributed to the large-scale increase in vegetation west of the Hu Line.

4.3 Policy implications of ecological restoration projects on vegetation change

We know that natural factors can promote the growth of vegetation. However, ecological restoration projects and vegetation restoration policies have played an increasingly important role in increasing vegetation cover (Chen et al., 2019; Piao et al., 2020). To better represent the persistence and extensiveness of anthropogenic interventions in vegetation change, we identified policies or ecological restoration projects that may have intervened in the growth of vegetation in the A1–A8 regions to explore their contributions to the overall trend of vegetation development. As shown in Table 4, the key ecological restoration projects and vegetation restoration policies implemented in China covered most of the regional vegetation in China. A series of measures have been implemented under several major ecological programs, including the Afforestation Program for Taihang Mountains, the Grain for Green Program, the Three-North Shelterbelt Program, the Natural Forest Protection Project, and the Program on Sandification Control for Areas in the Vicinity of Beijing and Tianjin. These measures encompass afforestation of bare land, conversion of cultivated land to forests and grasslands, grassland management, and grazing bans. In addition to national-level ecological restoration projects, some vegetation restoration projects, such as the Karst Ecological Restoration Project and the Ecological Restoration Program of Three-River Headwaters, have increased the area of vegetation restoration. In the future, the government should pay more attention to the restoration of grassland vegetation in ecologically fragile areas, such as the agricultural and pastoral zones on the west side of the Hu Line. Meanwhile, vegetation restoration efforts

Table 4 Major ecological restoration projects or policies related to vegetation restoration in China

Area	Duration	Vegetation trends	Program
A6, plain zones	1986–2050	Curbing land degradation and increasing vegetation cover (Feng et al., 2021)	Afforestation Program for Taihang Mountain
A7, A8	1989–2020	Increased vegetation cover (Qu et al., 2018)	Shelterbelt Program for Upper and Middle Reaches of Yangtze River; Shelterbelt Program for Pearl River
A1, A2, A3, A5, A6, A7, A8	1999–2020	Overall improvement in vegetation cover (Cao et al., 2009; Qian et al., 2019)	Grain for Green Program
A1, A3, A6	1978–2050	Increased vegetation cover and productivity (Ma et al., 2019; Zhai et al., 2023)	Three-North Shelterbelt Program
A2, A3, A4, A5, A6, A7	1998–2020	Significant area of natural forest revegetation (Wang et al., 2024)	Natural Forest Protection Project
A5	2005–2013	Substantial contribution to revegetation (Shao et al., 2017)	Phase I of the Ecological Restoration Program of Three-River Headwaters
A2, A6	2000–2022	Desertification trends improve and revegetation is effective (Niu et al., 2023)	Program on Sandification Control for Areas in the Vicinity of Beijing and Tianjin
A5	2000–2016	Significant increase in vegetation cover (Qiao et al., 2021)	Karst Ecological Restoration Project

Note: for detailed information on ecological restoration projects, please refer to The Encyclopedia of China (<https://www.zgbk.com>)

should prioritize cost-effective measures, as large-scale afforestation remains financially demanding and often leads to limited tree species diversity. Our research has confirmed that the rapid development of megacities has already had a negative impact on vegetation. Therefore, we suggest that future policies should also focus on the greening of vegetation in megacities to harmonize the impacts of natural factors on vegetation.

5 Conclusions

This study analyzed vegetation dynamics in eight regions of China from 2000 to 2020 and assessed the impacts of both natural and anthropogenic factors. The key findings are as follows: 1) between 2000 and 2020, vegetation coverage increased significantly across 44.14% of China ($P < 0.05$). Areas with reduced vegetation are mainly concentrated on farmland in central A6 and urban regions in eastern A7. Meanwhile, vegetation degradation on the west side of the Hu Line (A2, A3, and A4) improved markedly, with less than 1.00% of these areas exhibiting an extremely significant reduction ($P < 0.01$). 2) Vegetation changes in China are jointly driven by natural and anthropogenic factors through nonlinear interactions. Precipitation ($q = 0.766$) and land use type ($q = 0.636$) emerged as the most influential natural and anthropogenic factors, respectively, while their interaction ($q = 0.838$) dominated vegetation distribution patterns at the national scale. Regional responses were differentiated: on the west side of the Hu Line, vegetation dynam-

ics are primarily constrained by arid and semi-arid climates, with precipitation being the dominant natural factor. Ecological restoration projects have effectively increased vegetation cover in these regions, demonstrating the potential of human intervention to alleviate climatic limitations. On the east side of the Hu Line, vegetation growth benefits generally from warm and humid conditions, though localized decline in urbanized areas (e.g., A6 and A7) is largely attributable to land development and economic expansion. 3) The influence of precipitation strengthened over time in certain regions (e.g., q values increased by 26.73% in A3 and 101.13% in A4), reinforcing its role as a key driver of vegetation improvement on the west side of the Hu Line. Concurrently, anthropogenic factors, particularly land use type and population density, demonstrated increasing influence in eastern urban areas, as indicated by q -values rise of 37.22% in A2, 67.26% in A3, and 68.08% in A8. This study can help target efforts towards vegetation preservation in different regions in the future.

Supplementary

Figs. S1–S3 and Tables S1, S2 could be found at <http://egeoscienc.neigae.ac.cn>.

Conflict of Interest

The authors have no competing interests to declare that are relevant to the content of this article.

Author Contributions

LIU Yuan: conceptualization, methodology, formal analysis and investigation, writing-original draft preparation, writing-review and editing; WANG Chengyuan: software, writing-review and editing; WANG Enheng: writing-review & editing, supervision; MAO Xuegang: conceptualization, funding acquisition, supervision; FENG Tiantian: writing review and editing; YANG Liwei: software, investigation; NIU Qian: writing-review & editing; DING Longxin: software, investigation; SERGEY Chumachenko: writing-review & editing, supervision.

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